The Missing Institution: Solvable Demand as a Macroeconomic Function

A HANK and Two-Asset Evaluation for France

Sébastien Delestre
Independent Researcher
ORCID: 0009-0005-4654-5107

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Preliminary draft — This paper proposes a testable hypothesis. Results are illustrative, based on a calibrated model with documented limitations, and do not constitute policy recommendations. Comments, criticisms, and corrections are welcome.

Abstract

Modern macroeconomic debates remain trapped in a binary opposition between supply-side and demand-side approaches. This paper identifies a blind spot common to both: neither paradigm proposes an explicit, continuous institution dedicated to regulating the circulation of solvable demand. I propose Φ, a targeted automatic stabilizer that redirects €32 billion of existing tax expenditures into an hourly wage subsidy for 13 million low- and middle-income workers in France.

Using a one-asset Heterogeneous Agent New Keynesian (HANK) model calibrated to the French economy, I find that Φ generates a fiscal multiplier of 0.775 over eight quarters, within the empirical range of [0.5; 1.5] documented by Auerbach & Gorodnichenko (2012) and Ramey & Zubairy (2018). The policy lowers steady-state structural unemployment from 6.98% to 6.73%, and generates a net positive balance for the government of +€9.75 billion per year. The bulk of these gains comes from redirecting existing spending toward households with high marginal propensity to consume (MPC = 0.78), rather than from increasing the overall level of public spending. Φ does not generate significant inflation ($\Delta\pi = 0.197\%$).

To test the robustness of the mechanism in a framework where households face liquidity constraints and portfolio frictions, I extend the analysis to a Two-Asset model (Kaplan, Moll & Violante, 2018). The multiplier is structurally lower (0.199), but reveals three unexpected results. First, the selling friction (kink) is neutral for tax expenditure financing ($\Delta = 0.0055$). Second, the kink is crucial for lump-sum financing ($\Delta = +0.1222$): without it, Ricardian households overreact and the multiplier becomes negative. Third, the kink exhibits a bell curve with an economic optimum at $\chi_{0,sell} = 0.10$. To my knowledge, these mechanisms have not been explicitly identified in the fiscal multiplier literature.

Keywords: fiscal multiplier, HANK, Two-Asset, automatic stabilizer, solvable demand, tax expenditures, France.
JEL Codes: E62, E63, H24, H31.

Replication code, data, full documentation of the device (ARE, Sovereign Fund, transition plan), and figures are available on GitHub: https://github.com/sebastien-edl/phi-hank-france

1 Introduction

Modern macroeconomic architecture includes specialized institutions for several core functions: central banks for monetary stability, supervisory authorities for financial stability, antitrust bodies for competition, and welfare systems for social protection. No institution, however, is explicitly mandated to regulate the circulation of solvable demand — the purchasing power effectively available to convert needs into market transactions. This absence is not an error. It is a structural blind spot.

Supply-side approaches assume that demand emerges naturally from production and price adjustment. Demand-side approaches treat it as a policy variable, but only through discretionary intervention (stimulus, tax cuts), not through a permanent, rule-based institution. The result is a binary debate: supply or demand. Neither side proposes an institution dedicated to the continuous regulation of solvable demand itself.

This paper does not argue that demand determines everything. It argues that the institutional regulation of solvable demand is a missing function — a gap that neither paradigm has filled. I propose Φ, a candidate institution to fill that gap, and I test its effects in two models calibrated to France: a one-asset HANK model, and a Two-Asset model.

Contribution. This paper makes five contributions. First, it identifies a blind spot in the macroeconomic architecture. Second, it proposes a concrete candidate institution (Φ), with five properties (continuous, automatic, targeted, financed, reversible). Third, it quantifies the effects of Φ in a HANK model calibrated to France (multiplier of 0.775). Fourth, it extends the analysis to a Two-Asset model, which gives a more realistic multiplier (0.199) and reveals the crucial role of the selling friction. Fifth, it identifies three novel observations on the kink bell curve, the neutrality for niches, and the overreaction of Ricardian households without friction.

Results. The key results are summarized in Table 1.

Table 1: Main results (HANK and Two-Asset synthesis)[1]

Indicator	**HANK (One-Asset)**	**Two-Asset (Niches + Kink)**
Fiscal multiplier μ (8Q)	0.775	0.199
Empirical range (literature)	[0.5; 1.5]	[0.05; 0.30]
ΔY (annual, at peak)	+€27.51 bn	+€7.06 bn
Jobs created (Φ channel)	75.1k FTE	19.8k FTE
Structural unemployment (SS)	6.98% → 6.73%	7.48% → —[a]
Net government balance (central)	+€9.75 bn/year	+€9.41 bn/year
Inflation created ($\Delta\pi$)	0.197%	0.197%

Plan. Section 2 presents the missing institution. Section 3 describes the Φ device. Section 4 presents the two models. Section 5 reports the HANK results. Section 6 details the Two-Asset extension and the three observations. Section 7 discusses robustness and limitations. Section 8 concludes.

2 The Missing Institution

Modern macroeconomic architecture includes specialized institutions for several core functions:

- Monetary stability — central banks (ECB, Fed);

[1] Unemployment figures report steady-state comparisons (pre-Φ vs post-Φ equilibria), not transitional paths. The HANK model converges to the post-Φ steady state within approximately 8 quarters, consistent with the multiplier window used throughout. Transitional unemployment dynamics are left to a subsequent version of this paper. [a] The Two-Asset model does not generate a comparable transitional unemployment path; employment creation is reported in the Jobs row.

- Financial stability — supervisory authorities (ACPR, SEC);
- Competition — antitrust authorities;
- Social protection — welfare systems.

No institution, however, is explicitly mandated to regulate the circulation of solvable demand — the purchasing power effectively available to convert needs into market transactions. This absence is not an error. It is a structural blind spot.

3 The Φ Device

Φ is a targeted automatic stabilizer with five properties:

1. **Continuous** — permanent adjustment, not discretionary;
2. **Automatic** — rule-based, transparent algorithms;
3. **Targeted** — priority to agents with high marginal propensity to consume (MPC);
4. **Financed** — the subsidy is funded by redirecting existing tax expenditures, which neutralizes its net impact on unit labor costs for employers;
5. **Reversible** — automatic deactivation in case of overheating.

Table 2: Design of the Φ device

Component	**Value**
Base subsidy	€1.00 per hour worked
Normal surcharge ($\pi \leq 2\%$)	+€0.50/h
Vigilance surcharge ($2\% < \pi \leq 3.5\%$)	+€0.25/h
Galloping surcharge ($\pi > 3.5\%$)	€0.00/h
Recession surcharge	+€2.00/h base → +€2.50/h total (Sovereign Fund)
Target	Workers < €4,000 gross/month
Beneficiaries	13 million
Annual cost	€35.49 bn gross / €31.94 bn net

Financing. Φ is financed by redirecting €32 billion of existing tax expenditures. This is not a new tax. It is a redirection of existing spending toward agents with higher MPC. The redirection is progressive: in the first phase (years 1–2), only the least efficient tax expenditures are redirected; in the second phase (years 3–5), continued evaluation determines which expenditures are preserved; in the third phase (years 5+), only strategic expenditures (innovation, defense, ecological transition) are kept.

The decoupling of labor cost and purchasing power. A central point of the device deserves clarification: it introduces a *partial and institutional* decoupling between the labor cost borne by the employer and the employee's disposable income. In the current architecture, lowering labor costs requires lowering wages or payroll taxes, which reduces purchasing power; increasing purchasing power requires raising wages, which increases costs for the firm. Φ breaks this link: the hourly subsidy is paid to the worker under the public device, not under the employment contract. Simultaneously, the payroll tax cut financed by the net balance reduces costs without reducing gross wages.

Who bears the cost? The cost is redistributed, not eliminated. The €32 billion of redirected tax expenditures represent a benefit that some firms and wealthy households previously

enjoyed. By redirecting them to the Φ subsidy, the State transfers this cost from those who benefited from the niches to those who now benefit from it as purchasing power. For firms without niches, the device is neutral, or even positive (via the payroll tax cut). For firms benefiting from "comfort" or "brown" niches, the device represents a loss of this specific advantage, partially offset by the payroll tax cut and the increase in aggregate demand.

The institutional details of the device (the ARE countercyclical regulation mechanism with its five zones, the Sovereign Fund and its three accumulation scenarios, the three-phase transition plan) are documented in the full project documentation, available on GitHub: https://github.com/sebastien-edl/phi-hank-france

4 The Models: HANK and Two-Asset

4.1 The HANK model (one-asset)

The HANK model follows the tradition of Krusell & Smith (1998) and Kaplan et al. (2018). It is solved using the sequence-space Jacobian method of Auclert et al. (2021). The economy is populated by a continuum of households of measure one, indexed by their productivity level $e \in \{e_1, \ldots, e_{10}\}$ and their asset holdings a. There are ten states: five productivity levels for employed workers and five for unemployed workers.

Households solve:

$$\max_{\{c_t, a_t\}_{t \geq 0}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \tag{1}$$

subject to the sequential budget constraint:

$$c_t + a_t = (1 + r)a_{t-1} + z_t \tag{2}$$

where $z_t = w_t e_t + b \cdot \mathbb{K}\{e_t = 0\} + \phi_t - T_t$ is disposable income. The utility function is CRRA. The labor market follows the Krusell-Smith structure, with a job-finding rate that depends on aggregate output and on the payroll tax cut *bc*. The calibration gives $\beta_{FR} = 0.9862$, a HtM share of 26.94%, and an MPC of Φ beneficiaries of 0.7817.

4.2 The Two-Asset model (extension)

The one-asset HANK model assumes that all households hold the same form of savings. The Two-Asset model follows Kaplan, Moll & Violante (2018). Households hold a liquid asset b and an illiquid asset a. The portfolio is subject to frictions: selling the illiquid asset is costly. The key parameter is $\chi_{0,sell}$, which controls the intensity of the selling friction (the "kink"). The model is calibrated to French data with three statuses (E, U_short, U_long). It generates a HtM share of 26.09%, short-term unemployment of 5.63%, and long-term unemployment of 1.85% (total steady-state unemployment of 7.48%).

4.3 Financing of the Φ device

The gross annual cost of Φ is €35.49 bn. It corresponds to an hourly subsidy of €1.50/h paid to 13 million low- and middle-income workers, based on 151.67 monthly hours. After income tax (10%), the net cost is €31.94 bn. Financing comes from redirecting €32 bn of tax expenditures: the redistributive balance is almost zero (+€0.06 bn). The net balance of +€9.75 bn/year comes exclusively from macroeconomic effects (fiscal return on ΔY: +€8.18 bn/year; unemployment savings: +€1.51 bn/year).

Table 3: Decomposition of the net balance

Item	Amount (€ bn/year)	Nature
Redirected niches	+32.00	Revenue
Φ net disbursed	31.94	Expense
Redistributive balance	+0.06	Neutral
Fiscal return on ΔY	+8.18	Macro effect
Unemployment savings	+1.51	Macro effect
Central net balance	+9.75	Fiscal gain

5 HANK Results (The Main Channel)

The fiscal multiplier is computed over 8 quarters:

$$\mu_{8T} = 0.775 \tag{3}$$

This value is within the empirical range [0.5; 1.5]. The Φ channel creates 75.1k FTE, and the *bc* channel creates 293.6k. Steady-state unemployment declines from 6.98% to 6.73%. The net government balance is +€9.75 bn/year. The induced inflation is $\Delta\pi = 0.197\%$ (post-processing), below the 2% threshold.

6 Two-Asset Results and Novel Observations

The Two-Asset model introduces a selling friction. This section shows that it modifies the evaluation of financing modes, and isolates three mechanisms that, to my knowledge, have not been explicitly documented in the fiscal multiplier literature.

Table 4: Multiplier by financing mode and friction

Financing mode	With Kink ($\chi_{0,sell} = 0.10$)	Without Kink ($\chi_{0,sell} = 0$)
Niches (tax on the rich)	+0.199	+0.194
Broad (tax on everyone)	+0.237	—
Lump-sum (flat tax on the rich)	+0.044	-0.078

Observation 1: The neutrality of the friction for niches. The kink has almost no effect on financing through niches ($\mu = 0.199$ with kink vs $\mu = 0.194$ without kink, a difference of 0.005). Financing through niches does not incentivize Ricardian households to engage in massive asset sales. The multiplier is therefore structurally robust to the friction.

Observation 2: The crucial role of the friction for lump-sum. Without friction, lump-sum financing produces a negative multiplier ($\mu = -0.078$). With friction ($\mu = 0.044$), it becomes positive again. The lump-sum is a negative wealth shock. Without friction, Ricardian households overreact by selling their capital en masse (savings flight, demand collapse). The friction acts as a calming mechanism: it mechanically prevents overreaction.

Observation 3: The bell curve of the friction. The multiplier for lump-sum as a function of $\chi_{0,sell}$ shows a non-linear relationship (Table 5).

Table 5: Sensitivity of the multiplier to the selling friction (lump-sum)

Friction ($\chi_{0,sell}$)	**Multiplier** μ
0.00	-0.078
0.10	+0.044 (Optimum)
0.20	-0.070

There exists an economic optimum of friction ($\chi_{0,sell} \approx 0.10$). Below this threshold, friction is too low (overreaction). Above it, friction is too high (transaction blocking). This result, to my knowledge, had not been identified in the literature.

7 Robustness and Limitations

Stability tests. Four tests were conducted (Table 6). The model is stable, determined, and linear.

Table 6: Stability tests of the Two-Asset model

Test	**lump + kink**	**niches + kink**
Rank(A_0) = 5/5	Yes	Yes
min|det $J(z)$| > 0.01	0.0590	0.0597
$\|\lambda\|$ implicit < 1	0.8745	0.8745
Dispersion μ < 1e-3	1.02e-16	1.11e-16

Sensitivity to high friction. With $\chi_{0,sell} = 0.20$, the multiplier for niches remains positive ($\mu = 0.207$). The Φ mechanism does not depend on a fine calibration of the friction.

Documented limitations.

1. **One-asset HANK.** Underestimates HtM households (26.94% vs 30-35% INSEE). Results are conservative.
2. **Ad hoc coefficients.** The hiring elasticity η_{bc} and fiscal returns are calibrated by scenarios.
3. **Post-processed inflation.** Computed via $\pi = \kappa\hat{y}$, not simulated.
4. **Investment channel not modeled.** The effect of the Sovereign Fund on capital is not captured.
5. **Behavioral responses assumed.** Tax evasion and labor supply responses are not tested.
6. **Closed economy.** Import leakages are accounted for in HANK, but not in Two-Asset.
7. **Steady-state unemployment only.** The unemployment figures reported in this paper are steady-state comparisons, not transitional paths. A full characterization of the unemployment trajectory u_t (including potentially non-monotone adjustment dynamics) is left to a subsequent version.

The paper does not claim that Φ is optimal. It demonstrates that it is budgetarily sustainable, macroeconomically coherent, and more effective than the status quo at equal budget.

8 Conclusion

The binary opposition between supply-side and demand-side approaches has left a blind spot: no institution is explicitly mandated to regulate the circulation of solvable demand. Φ is a candidate institution to fill that gap. Tested in a HANK model, it generates a multiplier of 0.775, creates 369,000 jobs (75,100 via Φ, 293,600 via the payroll tax cut), lowers steady-state unemployment from 6.98% to 6.73%, and produces a net balance of +€9.75 bn/year without significant inflation.

The Two-Asset extension confirms the robustness of the mechanism and reveals three unexpected results. First, the kink is neutral for niches. Second, it is crucial for lump-sum: without it, Ricardian households overreact and the multiplier becomes negative. Third, there exists a bell curve of the friction, with an economic optimum at $\chi_{0,sell} = 0.10$. To my knowledge, these mechanisms have not been explicitly documented in the literature.

A subsequent version will extend the analysis to transitional unemployment dynamics and the full u_t path, which the current steady-state comparison deliberately abstracts from.

The paper does not claim to have found the truth. It proposes a testable hypothesis, documents its limitations, and invites criticism. The code, data, and full documentation of the device are available on GitHub: https://github.com/sebastien-edl/phi-hank-france

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A INSEE Data (2024 T3)

Table 7: INSEE ratios (2024 T3)

Ratio	**Value**
C/Y (consumption / GDP)	50.6%
I/Y (investment / GDP)	21.6%
M/Y (imports / GDP)	31.4%
X/Y (exports / GDP)	31.8%

B AI Disclosure and Author Note

AI Usage Disclosure. The author used AI assistants (including for Python code generation and drafting assistance) in the preparation of this work. All numerical results were verified, recomputed, and validated by the author. The author takes full responsibility for the content, any remaining errors, and the conclusions.

Note on the Author. Sébastien Delestre is a self-taught researcher in economics, without formal academic affiliation or degree in the field. This work was carried out outside any academic institution, with the means available. The author has done his best to make this work understandable to professional economists, but he does not claim their level of knowledge or technical competence. He proposes an idea — the institutional regulation of solvable demand — which, if judged worthy of interest, deserves to be studied, deepened, criticized, or refuted by professional researchers. The author accepts in advance that this idea should be buried if it does not withstand scrutiny. This work was carried out with the greatest possible impartiality. Any remaining errors are not intended to conceal anything or to mislead the reader; they reflect the limits of the author's means and knowledge. Corrections and criticisms are welcome.